

1) Apply all the CPU scheduling algorithms for inscm examination of OSCP.

	AT	BT	Priority
P ₁	0	4	3
P ₂	0	9	1
P ₃	0	.6	2
P ₄	0	4	3
P ₅	0	9	1
P ₆	0	6	2
P ₇	1	4	3
P ₈	1	9	1
P ₉	1	6	2

2) Define following terms

* Starvation

* Convoy effect

* Context-switching

* Aging.

3. A Mobile has the following tasks
draw a table with

Task	AT	BT
Music Player	0	20
Touch even	3	1
WhatsApp	5	2
Camera	6	4
Launch		

- i) which scheduling algorithm is best suitable for above problem.
 - ii) Draw the Gantt chart for SRTF
 - iii) why selected algorithm for improve user experience.
 - iv) which task receives the fastest response.
- 4) Solve the following problem with round robin algorithm time quantum of $R = 1$.
Context-switch time of 0.5. Draw a table

Process	AT	BT
P_1	0	5
P_2	0	4
P_3	0	3

Solution

1) FCFS $\rightarrow$ First Come First Serve

Ready Queue
Arrival time

P ₁	P ₂	P ₃	P ₄	P ₅	P ₆	P ₇	P ₈	P ₉
0	0	0	0	0	0	1	1	1

Gantt chart

P_1	P_2	P_3	P_4	P_5	P_6
0	4	13	19	23	32

38

P ₇	P ₈	P ₉
42	51	57

$$TAT = CT - AT$$

$$WT = TAT - BT$$

$$= CT - AT - BT$$

	AT	BT	CT	TAT	WT
P ₁	0	4	4	4	0
P ₂	0	9	13	13	4
P ₃	0	6	19	19	13
P ₄	0	4	23	22	19
P ₅	0	9	32	32	23
P ₆	0	4	38	38	32
P ₇	1	4	42	41	37
P ₈	1	9	51	50	41
P ₉	1	6	57	56	50
				<u>276</u>	<u>217</u>

$$\text{Avg. TAT} = \frac{276}{9} = 30.66$$

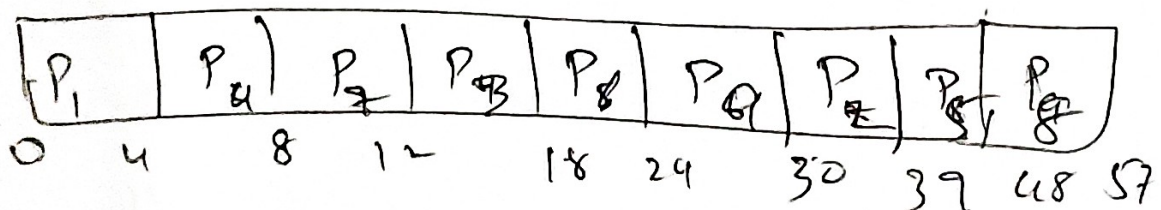
$$\text{Avg. WT} = \frac{219}{9} = 24.33$$

SJN (Shortest Job Next) / Shortest Job first (SJE)

Process	AT	BT	Process	Process	AT	BT
P ₁	0	4	-	P ₆	6	1
P ₂	0	9		P ₇	1	4
P ₃	0	6		P ₈	1	9
P ₄	0	4		P ₉	1	6
P ₅	0	9				

i. Shortest BT = 4

Gantt chart



3h
BT
6
4
8
9
6

	AT	BT	CT	TAT	WT
	0	4	4	4	0
P ₁	0	4	8	8	4
P ₂	1	4	12	11	7
P ₃	0	6	18	18	12
P ₄	0	6	24	24	18
P ₅	0	6	30	29	23
P ₆	1		39	39	30
P ₇	0	9	49	48	39
P ₈	0	9		56	47
P ₉	1	9	57	<u>237</u>	<u>180</u>

$$\text{Avg TAT} = \frac{237}{9} = 26.33$$

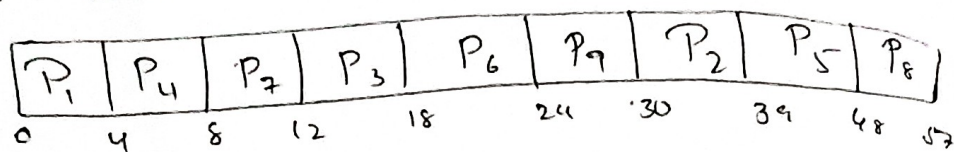
$$\text{Avg WT} = \frac{180}{9} = 20$$

SRTF:- (Shortest Remaining Time first)

At time 0 P₁ start RBT = 4 - 1 = 3 at
time 1 P₇ P₈ P₉ arrive

P ₁ 3	P ₉ 6
P ₄ 4	P ₂ 4
P ₇ 4	P ₅ 9
P ₃ 6	P ₈ 9
P ₆ 6	

Gantt chart



Same values as SJN.

Priority Scheduling - (Non-Preemptive)

Highest Priority = 1

Priority - 2

P₂ → P₅

27 → 33 : P₃

0 → 9 : P₂

33 → 39 : P₆

9 → 18 : P₅

39 → 45 : P₉

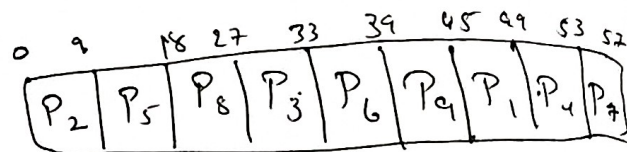
18 → 27 : P₈

Priority 3:-

45 → 49 : P₁

49 → 53 : P₄

53 → 57 : P₇



Gantt chart

	CT	TAT	WT
P ₁	49	49	45
P ₂	9	9	0
P ₃	33	33	27
P ₄	53	53	49
P ₅	18	18	9
P ₆	39	39	33

CT TAT WT

P₇ 57 56 52

P₈ 27 26 17

P₉ 45 $\frac{44}{327}$ $\frac{34}{270}$

$$\text{Avg. TAT} = \frac{327}{9} = 36.3$$

$$\text{Avg. WT} = \frac{270}{9}$$

Priority Scheduling

At time 0, P₂

0 → 9 : P₂

0 → 9 : P₂

9 → 18

18 → 27

Priority - 3:-

45 → 49 : P₁

49 → 53 : P₄

53 → 57 : P₇

So, star. thi

2) i) Starvation

occurs

or for an

other process

execution ..

Preemptive

depending

Priority Scheduling (Preemptive)

At time 0, P_2 has Priority 1. Priority 2:-

0 $\rightarrow$ 1 : P_2

27 $\rightarrow$ 33 : P_3

0 $\rightarrow$ 9 : P_2

33 $\rightarrow$ 39 : P_2

9 $\rightarrow$ 18 : P_5

39 $\rightarrow$ 45 : P_9

18 $\rightarrow$ 27 : P_8

Priority - 3:-

45 $\rightarrow$ 49 : P_1

49 $\rightarrow$ 53 : P_4

53 $\rightarrow$ 57 : P_7

So, for this particular data

Preemptive = Non-Preemptive.

4) i) Starvation:-

occurs when a process waits indefinitely or for an excessively long time because other processes are repeatedly selected for execution. Starvation can occur in both Preemptive and Non-preemptive scheduling depending on the algorithm.

ii) Convoy effect

CE is a important problem associated with non-preemptive scheduling especially FCFS. The long process ask like a convoy, causing many short process to follow behind it.

iii) Context Switching.

CS is process of saving the state of the currently running process and loading the saved state of another process so that the CPU can switch b/w them.

iv) Aging:-

A is a technique used to prevent starvation by gradually increasing the priority of a process that has been waiting for a long time.

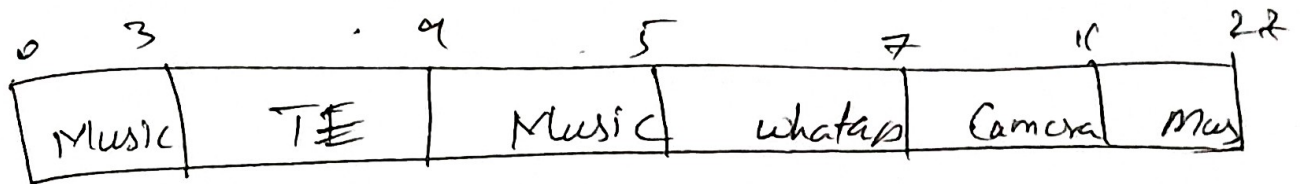
3) i) SRTF always select the process with the shortest remaining Burst time.

ii) At time 0: only Music Player
music plays has executed for 3 units

$$RBT = 20 - 3 = 17$$

At time 3: Touch Event arrives

	RBT
Music player	17
Touch Event	1

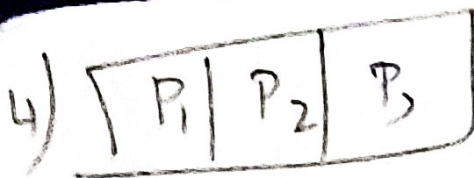


iii) SRTF improves the user experience bcz it gives short interaction tasks priority over long CPU intensive tasks.

iv) RT = First CPU start time - At

	AT	FS	RT
Music	0	0	0
Touch	3	3	0
whatsapp	5	5	0
Camera	6	7	1

The fastest Responses time are music plays, Touch Event, whatsapp notify.



* P₁ requires 5ms

$$P_1 : 0 \rightarrow 2$$

$$\text{Remaining} : 5 - 2 = 3\text{ms}$$

$$\text{Context-switch} = 2 \rightarrow 2.5$$

$$P_2 \rightarrow P_3 \rightarrow P_1$$

* P₂ required 4

$$P_2 : 2.5 \rightarrow 4.5$$

$$\text{remaining} : 4 - 2 = 2$$

$$P_3 \rightarrow P_1 \rightarrow P_2$$

$$\text{Context switch} : 4.5 \rightarrow 5$$

* P₃ requires 3ms

$$P_3 : 5 \rightarrow 7$$

$$3 - 2 = 1$$

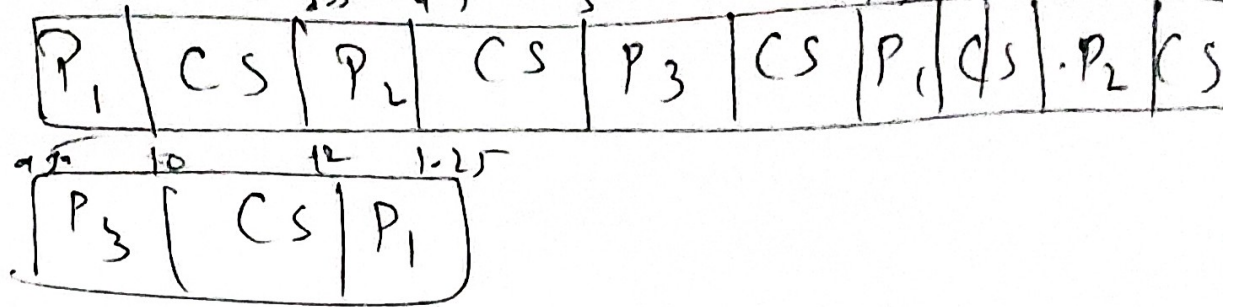
$$P_1 \rightarrow P_2 \rightarrow P_3$$

* P₁ has 3ms remaining

$$P_1 : 7.5 \rightarrow 9.5$$

$$3 - 2 = 1$$

$$9.5 \rightarrow 10.5 \rightarrow P_2 \rightarrow P_3 \rightarrow P_1$$



P ₁	CT 15
P ₂	12
P ₃	13.5

$$TAT = CT - AT$$

	TAT	WT
P ₁	15	10
P ₂	12	6
P ₃	13.5	10.5

Avg wait = 9.5

Avg TAT = 12.5